HDOC - Day 12

Menu-Driven C Program Using Switch-Case

Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum and product of the first and last digits.
- Count how many digits are less than, equal to, and greater than a given digit k.

Algorithm

1. Start.

2. Input a positive integer num.

3. If num <= 0, display "Please enter a positive integer" and stop.

4. Display the menu: 1. Sum and product of first and last digits 2. Compare digits with k.

5. Input the user's choice.

6. Use switch(choice).

7. Case 1: Set last = num % 10. Set temp = num. Repeatedly divide temp by 10 while temp >= 10. The remaining value is the first digit. Calculate sum = first + last and product = first * last. Display the results.

8. Case 2: Input digit k. If k is outside 0 to 9, display an error. Otherwise set lesser = equal = greater = 0 and temp = num. Extract each digit using temp % 10. Compare it with k and increment the appropriate counter. Divide temp by 10 and repeat until temp becomes 0. Display all three counts.

9. Default: Display "Invalid choice".

10. Stop.

Test Case Table

Test	Number	Choice	k	Expected Output
1	12345	1	-	Sum = 6, Product = 5
2	12345	2	3	Lesser = 2, Equal = 1, Greater = 2
3	123	3	-	Invalid choice
4	-12	-	-	Please enter a positive integer

C Program

```
#include <stdio.h>

int main()
```

```c
{
    int num, choice, temp, first, last;
    int sum, product, k, digit;
    int lesser = 0, equal = 0, greater = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Sum and product of first and last digits\n");
    printf("2. Count digits lesser than, equal to and greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            last = num % 10;
            temp = num;

            while (temp >= 10)
                temp = temp / 10;

            first = temp;
            sum = first + last;
            product = first * last;

            printf("Sum = %d\n", sum);
            printf("Product = %d\n", product);
            break;

        case 2:
            printf("Enter digit k (0-9): ");
            scanf("%d", &k);

            if (k < 0 || k > 9)
            {
                printf("Please enter k between 0 and 9.\n");
                break;
            }

            temp = num;
            while (temp > 0)
            {
                digit = temp % 10;

                if (digit < k)
                    lesser++;
                else if (digit == k)
                    equal++;
                else
                    greater++;

                temp = temp / 10;
            }
```

```
            printf("Count_lesser = %d\n", lesser);
            printf("Count_equal = %d\n", equal);
            printf("Count_greater = %d\n", greater);
            break;

        default:
            printf("Invalid choice.\n");
    }

    return 0;
}
```

Compilation and Execution

Compilation command: gcc num.c -o num

Execution command: ./num

Compilation

```
┌──(kali㉿kali)-[~/day12]
└─$ gcc num.c -o num
```

Execution - Choice 1

```
┌──(kali㉿kali)-[~/day12]
└─$ ./num
Enter a positive integer: 12345

MENU
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 1
First digit = 1
Last digit = 5
Sum = 6
Product = 5
```

Execution - Choice 2

```
┌──(kali㉿kali)-[~/day12]
└─$ ./num
Enter a positive integer: 12345

MENU
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 2
Enter digit k: 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2
```

Execution - Invalid Choice

```
┌──(kali㉿kali)-[~/day12]
└─$ ./num
Enter a positive integer: 123

MENU
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 3
Invalid choice.
```

Execution - Invalid Positive Integer

```
┌──(kali㉿kali)-[~/day12]
└─$ ./num
Enter a positive integer: -12
Please enter a positive integer.
```